

## MSc Project - Reflective Essay

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| <b>Project Title:</b>      | A Comparative Study for Machine Learning and Deep Learning Approaches for Cybersecurity Intrusion Detection System |
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This paper studies the various state-of-the-art techniques that have been used in order to build an intrusion detection system in this digital age. There have been numerous accounts of cyberattacks in the recent times. The data leaks around the world have left a lot of people in complete exposure with their private and confidential data out in the open, publicly. As a result of this, many people are likely targets to huge number of scams, for example, scams through various phishing techniques, whether it is operated via emails, voice calls or anything else. These attacks are not only affecting people through different scams or data leakage, but also affecting people's life. There have been a few incidents where people's lives have been endangered because of a few cyberattacks which adversely also had some counter effects on lives of people, where even death has been one of the tragic outcomes that is seen indirectly due to disturbance caused during a cyberattack. Losing lives in a cyberattack signifies that it is not only related to our digital life and digital gadgets, but also that as our life revolves around technology for most of the time, these cyberattacks have the capability to cause imbalance and bring chaos into our lives, as an individual and also as a global society. Thus, there is a significant need to take the necessary steps in improving our security infrastructure and make it as impenetrable for hackers as possible. However, as technology evolves day-by-day, it seems obvious that no matter how strong and secure an infrastructure is built, evil minds being creative would invent new techniques and software to cause harm to the digital society. Thus, one needs to be ready for situations where the infrastructure fails to protect the entire system and respective measures need to be placed in terms of an event of an adverse situation, as an incident response. Penetration testing, risk mitigation and a few other steps are considered for this and are in practice these days. One such method also describes building an intrusion detection system and leave it in place to secure the system.

The main use of an intrusion detection system, also commonly known as IDS, is that it monitors the system network and connections constantly. It is usually trained upon normal network traffic behaviour for the system and is designed to spot abnormalities in the flow of network. It studies the network traffic and takes the necessary precautions when it comes across any anomalies in the traffic and suspects any possibility of compromising the system.

Initially, the study started with a comparative study of various research papers and different surveys about the works towards prevention of cybersecurity attacks, it can be seen that most of the researches conducted used different datasets like KDD CUP 1999, NSL KD'99, Kyoto 2006+, UNSW NB15, ISCX-2012, NSL KDD, CIDD5001, CICIDS2017 etc. However, the majority of the work has been conducted using the NSL KDD dataset with all papers using different machine learning and deep learning approaches by using Neural Networks, Restricted Boltzmann Machines, Logistic Regression, Stochastic Gradient Descent, Adaptive Boosting, Random Forest, Naïve Bayes method and a plenty of other different methods.

However, as the CSE-CIC-IDS 2018 dataset is a very recent dataset, very less research work has been carried out upon it when compared to the other datasets. Also, as hackers find new ways of getting through the existing secure systems and evade all the security measures put into place, new updated measures need to be taken with the latest state-of-the-art dataset. Thus, the CSE-CIC-IDS dataset was artificially prepared. It was under the joint venture of the Communications Security Establishment (CSE) and the Canadian Institute of Cybersecurity (CIC)<sup>1</sup>. The main reason behind creating it was to use it for building state-of-the-art security measures. It is usually seen that access to new or updated security datasets is very limited and sometimes not even shared due to privacy concerns. This then affects the future research work adversely, as one needs credible and latest data to build latest security measures and without it, the digital Fort cannot be built and held strong.

The current dataset in use, CSE-CIC-IDS 2018 dataset contains 10 individual datasets with most of them having more than 1 million records, with each describing network traffic for a specific day and details about specific attacks according to the days. As seen in a few existing researches on this dataset, a model is created, trained and tested for each type of dataset to create a respective intrusion detection system. However, it is very unrealistic to use all different models simultaneously in real life as before any attack it is almost impossible to know if there is going to be an attack, and of which kind, and thus, different models for detecting the anomalies do not seem to be used as a realistic approach to build an intrusion detection system. Thus in this paper, data of all different kinds attacks from all the individual datasets is taken into account and one generic model that fits well enough to predict any and all kinds of cyberattacks as described in the datasets for the intrusion detection system.

The beginning of the work on our datasets started with the pre-processing of our dataset. Here, the respective machine learning and deep learning models on the first data set by splitting it into training and validation datasets. Upon training the model on these datasets, it was then tested on all the other nine remaining data files. While the model provided an astonishing accuracy of more than 99 percentage for the training and validation datasets, it failed miserably in the testing phases. Also, it did not just fail on one dataset, but it failed drastically on almost all the datasets. It provided a huge disappointing result in the accuracies for the test datasets varying from 30 percentage to 80 percentage. This is totally unacceptable from a machine learning or a deep learning model as even a blind guess gives a 50% accuracy of detecting if the network traffic is harmful or not. This was a huge shock because it was very unclear about the reasons for such drastic variations and poor performance in the testing accuracies for all different datasets. Upon careful inspection and analysis of all the individual data sets, it was found that the main error was that, while processing of all the datasets, the labels in all of them were classified as benign or harmful type of cyberattacks, for all the datasets irrespective of the type of attacks described in it. This blind generalization of datasets and training a model on one dataset and testing it on a completely different kind of dataset, was the main reason for such a miserably failing model. This can also be seen in the explanatory data analysis of the datasets that the first data set includes network traffic data for either benign, and FTP or SSH brute force attacks. However, the rest of the other datasets describe cyberattacks like DDoS, DoS, botnet, SQL injection and other types of attacks. Also, the size of the data set available is huge as there are more than 16 million records in all 10 datasets collectively, describing the different kinds of network traffic for the respective cyberattacks. Moreover, the number of records describing normal network traffic i.e. Benign, is very high when compared to the various types of harmful network traffic described as various attacks. Thus, in order to make sure that the machine learning and deep learning models do not bias over the Benign attacks due to a huge disproportion of unequal number of samples, the number of samples for the benign

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<sup>1</sup> Available from <https://www.unb.ca/cic/datasets/ids-2018.html> [online]

attacks were reduced such that they were almost equivalent to the number of harmful network traffic. Even after these operations on the dataset files, it was still very big in size.

As a result of this huge size of dataset, it was too heavy for the respective GPU to process all the various operations on the datasets needed for this project, resulting in a very slow process. This was also one of the reasons due to which the dataset was trimmed down in size. Thus, it was decided to work on the entire dataset individually as provided in different files and so, various functions were created in order to operate them on all dataset files one by one according to the type of data filled in. It was also noted that all the individual data frames created from the respective data files had a similar kind of data structure with same columns/features included in them and it was seen that none of them were misaligned with each other because if it did, then it would obstruct the creation and training of our machine learning or deep learning model. On performing the aforementioned operations, it was found that the fourth data frame 'df4' created from the file '02-20-2018.csv' had a different shape when compared than the other data frames. Upon careful inspection, it was seen that there were four extra features in the fourth data frame which were relatively unnecessary for the training of our data and so, they were dropped from the data frame as none of the nine remaining data frames did not contain any of those features.

Furthermore, it was encountered that a few records in the eight and the ninth data frame, 'df8' and 'df9', had a few attribute values which exceeded the upper limit for a float data type, which obstructed further pre-processing of the data. As the amount of data available is abundant, dropping a few rows would not impact the models that would be trained for this research. Therefore, the records with those values exceeding that limit were also dropped from the data frames and thus, resolving the respective errors faced during the pre-processing of all the data frames without any loss of important information from any of the data frames.

## Confusion matrix and other metrics of calculation:

After training and testing the various models as described in the paper, the accuracy and confusion matrix metrics are used respectively for all models to compare them from one another. Below mentioned are the findings for all our models:

| Model                       | TP     | FP    | FN    | TN     | F1 Score | Accuracy |
|-----------------------------|--------|-------|-------|--------|----------|----------|
| Feed Forward Neural Network | 405325 | 1498  | 28006 | 442789 | 0.97     | 96.64%   |
| Logistic Regression         | 373769 | 33054 | 27238 | 443557 | 0.94     | 93.13%   |
| Decision Tree Classifier    | 392179 | 14644 | 19923 | 450872 | 0.96     | 96.06%   |

|                           |        |      |      |        |      |        |
|---------------------------|--------|------|------|--------|------|--------|
| Support Vector Classifier | 157375 | 5392 | 8441 | 179839 | 0.96 | 96.06% |
|---------------------------|--------|------|------|--------|------|--------|

## Strengths and weaknesses:

The main strength of this project is that one model has been created for all types of cyberattacks which on comparing with other research conducted on the CSE-CIC-IDS 2018 dataset, it is seen that various models are trained individually for different cyberattacks. Upon careful investigation, one can say that around five to six Machine Learning or Deep Learning models would be required for the different types of attacks in the current dataset. However, implementing so many models simultaneously for an intrusion detection system seems realistically very cynical as it is tough to predict which model is supposed to be used in which scenario and then using that prediction, one must predict if the network traffic might be harmful or not. This seems very sceptical as Hackers nowadays seem to use plenty of techniques to fool the system and without any prior knowledge about the type of attack, the use of these kind of intrusion detection systems become very limited. Considering this instance, this research discusses a generalised intrusion detection system to detect all kinds of cyberattacks according to the data available in the dataset in discussion. Also, when training a model without correlation and other pre-processing operations, it is observed that the dataset takes a large amount of time to train the model. Nevertheless, the accuracy provided by this model is very close to perfection ~ 99% upon the training and validation data. But upon deeper investigation into this, it can be seen that this specific model predictions contains a lot of true and false positives, whereas the number of true or false negatives predicted is found to be extremely low. This means that it predicts almost everything as positive i.e. benign attacks because the dataset used to train the model without pre-processing has extremely high number of samples of benign attacks on comparison to the other harmful attacks. Even though the accuracy provided by the model seems extremely good, it needs to be pointed out that it is just an illusion. it is too risky to use in real time as it assumes that almost the entire network traffic is benign and not malicious/harmful and thus, is more vulnerable to cyberattacks with the illusion of security provided by the intrusion detection system using this model without pre-processed data.

However, the best and most accurate model using pre-processed data in this research paper provides an accuracy of almost 97 percentage i.e. there is still a risk of three percentage where our model fails to detect a harmful signal in the network flow traffic resulting in the penetration of our intrusion detection system and thus, leading to a successful cyberattack which might wreak a havoc for the system in place. The damage and the related repercussions due to a cyberattack cannot be undermined as all cyberattacks are a threat to people, companies and government, and their data rights and privacy and other loss and risks.

Nevertheless, if more time had been available for this research, a more systematic and a better approach could have been implemented where it could have been possible to implement this developed network-based intrusion detection system in a real time environment and develop it accordingly further. Moreover, the system around it could have been developed such that instantaneous emergency precautions would be taken so that little to no harm is done to our system through triggering alarms to alert our system and other measure. Also, steps can be taken which can thus provide help in real time to counterattack the malicious intruder with dangerous motives, helping us to fight back the cyber criminals. Moreover, in the instance where our intrusion detection system has failed and the system is under attack, measures can be taken such as alerting the

Incident Management teams about the situation, shutting down the infected or compromised part of the system by itself so that no further damage is done to the important assets in discussion.